

The Vulnerability of Privacy Protection in Chinese Digital Society: A Critical Review through Two Cases

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China's emerging data privacy framework presents a significant paradox, diverging from Western models by embedding privacy within a state-centric vision of digital governance. While initiatives like the online real-name system and the mandatory disclosure of IP locations have already compromised user privacy, the state strategically frames its interventions as protective. The latest example is the implementation of the Cyber Trusted Identity ("CTID") system, which allegedly aims at enhancing individual privacy protection and combating online fraud but has sparked much controversy. This paper critically examines this tension, arguing that individual privacy in China's digital society is systematically subordinated to broader state governance objectives. Through qualitative legal analysis and in-depth representative case studies of the Didi cybersecurity review and the implemented CTID system, it investigates how privacy is strategically instrumentalized to achieve dominant broader digital governance objectives, such as data sovereignty strategy and digital surveillance in China's Internet society. The study reveals that the vulnerability of privacy is amplified by key factors like the selective enforcement of data security laws to prioritize state data sovereignty, legal ambiguities that legitimize expanded state oversight, and the deployment of centralized technological tools like the CTID, which solidifies state control over citizens' online activities.

Keywords: Cyber ID, digital governance, digital law, digital society, digital surveillance, privacy protection

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1. Introduction

With the largest digital populations in the world, China confronts the enormous challenge of governing an online ecosystem shaped by hundreds of millions of active internet users (Jaguścik & Imbach, 2022). While the European Union (EU) and United States (US) have established distinct models for privacy protection, China has charted a divergent path, often described as “data privacy with Chinese characteristics” (Pernot-Leplay, 2020). This divergence is rooted in a distinct socio-political context and geopolitical ambitions, manifesting in legislative objectives that explicitly prioritize national security and social order—rationales less prominent in Western data protection frameworks (Pernot-Leplay, 2020). In this model, digital technology has been a toolkit for the government to tighten social control, rather than streamlining administrative procedures (Lee, 2019; Hu & Zhang, 2023).

A prominent and recent manifestation of this distinct governance model is the promulgation of the *Measures on the Management of the National Online Identity Authentication Public Service* (CTID Measures) on 23 May 2025 by six authorities under Chinese central government (Ministry of Public Security et al., 2025), including the Ministry of Public Security (MPS) and the Cyberspace Administration of China (CAC). This initiative, actualizing the proposed CTID system initially outlined in a July 2024 draft for comment, establishes a nationally unified online identity authentication credential. The CTID system, already trialed by over 80 applications including ubiquitous platforms such as WeChat, Rednote, and Taobao (Chengdu Commercial Daily, 2024), aims to verify the real identity information of natural persons in cyberspace. However, the CTID system intensified public debates surrounding individual privacy and digital rights in the context of state digital governance (BBC, 2024).

The CTID system is not an isolated phenomenon but rather the culmination of a consistent legislative and regulatory trajectory aimed at diminishing anonymity and augmenting state oversight within China’s online sphere. This trend commenced notably with the 2016 Cybersecurity Law (CSL), which mandated real-name online registration (National People’s Congress, 2016). Subsequent measures, including the nationwide implementation of health codes for pandemic prevention and control in 2020 (CAC, 2020), the 2022 CAC provisions requiring the display of user IP address locations (CAC, 2022c), and the 2023 CAC notice strengthening “Personal Media” management (CAC, 2023), progressively tightened controls. Further, in 2023, major Chinese online platforms such as TikTok, WeChat, and Weibo voluntarily compelled users with substantial follower counts to publicly display their real names (Tu, 2024). Collectively, these actions underscore a persistent governmental drive towards the erosion of cyberspace anonymity (Lee & Liu, 2016, Yang, 2023), the cession of personal privacy, and the consolidation of control over personal data, with the CTID system representing a significant institutionalization of this approach.

Against this backdrop, this paper interrogates the characteristics of China’s privacy protection framework and explores the mechanisms through which users’ online privacy become vulnerable despite China’s increasingly systematic data protection law. This paper employs the theory of “authoritarian privacy” of Jia (2024) as an analytical lens, a concept that resonates with the aforementioned unique approach, into a critical analysis of two representative cases in China’s digital society, *i.e.*, the cybersecurity-related regulatory actions against Didi Global Inc. and the recent implementation of the CTID system. Through ambiguous legislative clauses permitting government enforcement expansion, the state’s portrayal of itself as a benign protector, and selective enforcement practices, the vulnerability of individual online privacy is

systematically magnified. However, these strategies serve to normalize state digital surveillance and privacy compromises, while mitigating dissent. The paper begins by reviewing the state-centric nature of privacy within China's digital governance and the resulting *de facto* vulnerability of individual online privacy. It then outlines the case study approach and the qualitative content analysis methodology used to examine how the narrative of privacy protection is strategically utilized in digital laws and enforcement practices. Finally, the findings of each case are discussed, demonstrating the discourse of personal privacy protection is strategically employed to justify broader digital governance ambitions, most notably data sovereignty and digital surveillance, while ultimately rendering individuals more exposed and less empowered.

2. State-Centric Privacy in China's Digital Society Governance

China's existing data protection framework, marked by key pillars like the CSL, the Data Security Law (DSL) and the Personal Information Protection Law (PIPL), articulates several objectives: safeguarding national security and cyberspace sovereignty, guarding the rights and interests of individuals' personal data, and promoting economic growth. However, He (2023) argued that the distribution of emphasis among these goals is perceived as unbalanced, with personal privacy often bearing the cost. Therefore, a mere recital of these objectives offers an incomplete understanding of China's distinctive legislative approach and its practical consequences (Jia, 2024), which underscores the need for clarifying the rationale underpinning China's unique path in privacy protection.

A consensus within the literature acknowledges fundamental divergences between Chinese and Western conceptions of privacy protection (De Hert & Papakonstantinou, 2015; Roberts, 2022; Jia, 2024; Zhang, 2024a). Rather than a straightforward transposition of established Western models, China's privacy regime has evolved organically, shaped by its distinct socio-political context and economic ambitions (Pernot-Leplay, 2020). This context informs a legal culture wherein concerns about online dissent and activities perceived as detrimental to social stability significantly influence regulatory design (Lee & Liu, 2016). Consequently, privacy is not conceptualized as an absolute personal right but rather as a constituent of a reciprocal arrangement of obligations between the state and citizens. As Roberts (2022) points out, within this paradigm, state-led digital surveillance is often deemed tolerable, provided it serves overarching interests of national security and social order. Illustratively, Article 12 and 28 of the CSL impose affirmative duties on individuals and organizations to "observe public order and respect social morality" and to "provide technical support and assistance to public security organs and state security organs", thereby signaling the subordination of individual privacy claims to state governance needs. To understand the mechanisms that produce the vulnerabilities explored in this paper, this study adopts the theoretical framework of "authoritarian privacy" as articulated by Jia (2024). He posits that in non-democratic contexts (such as China), privacy legislation, while ostensibly protecting individuals, primarily functions to seek enhanced control and bolster authoritarian stability. This is achieved not by genuinely empowering citizens with robust privacy rights enforceable against the state, but by managing popular discontent over data exploitation through legal and symbolic gestures, thereby legitimizing state control (Jia, 2024; Wang et al., 2024). Lee (2019) has noted that authoritarian regimes employ datafication and dataveillance not merely for administrative efficiency, but as core instruments of governance, turning technological infrastructures into tools of political control. This paper leverages Jia's framework not to engage in a broader

discussion of regime legitimacy, but specifically to analyze how privacy-related legal instruments are deployed to enhance state oversight and social control, rendering individual privacy relatively vulnerable in interactions with state power.

The existing literature further provides more evidence for the main points of “authoritarian privacy” by identifying several distinct characteristics of China’s privacy governance. The first is rule by law rather than rule of law. That is, while legal frameworks exist, they are often designed to serve state interests rather than to impose enforceable constraints on state data collection behavior. A significant legislative lacuna exists concerning robust and enforceable limitations on governmental data collection, processing, and utilization (Li et al., 2017; Pernot-Leplay, 2020; Gorman, 2021; Creemers, 2022). A conceptual separation is evident in Chinese law, distinguishing privacy protections applicable to private sector actors from those concerning state actions (Pernot-Leplay, 2020), while the relationship between state actions and individual privacy is not typically governed in a strict manner. Li et al. (2017) find that, for example, the CSL’s privacy protections are interpreted as not seemingly extending to governmental conduct. Similarly, the PIPL, while comprehensive in its regulation of data processing by both private entities and public organizations, imposes markedly less stringent constraints on state organs (Creemers, 2022). The DSL, in turn, does not aim to define the horizontal rights and obligations between individuals and data controllers, but rather being primarily oriented towards mitigating data-related risks to state integrity and national security, particularly those detrimental to party-state integrity (Creemers, 2022). Even where China’s data legislation incorporates provisions addressing data processing activities by state agencies, these are often accompanied by broad derogations that allow to legitimize extensive state incursions (He, 2023; Jia, 2024; Miao, 2024). For instance, Article 35 of PIPL exempts state organs from notification obligations when processing personal data under certain vaguely defined circumstances. Notably, such broad exception clauses for state organs are not uncommon throughout Chinese legislative drafting, mirrored in provisions such as Article 41 of the DSL (state organs may not disclose government data) and Article 1035 of the Civil Code (personal information can be processed without consent).

The second is the prevalence of rule by policy rather than rule by law. China’s operationalization of privacy-related controls is frequently facilitated through a diverse array of government-issued documents, irrespective of their formal standing within the normative hierarchy. This reflects an administrative ethos often characterized as “rule by policy” (Miao et al., 2021). Zhang (2024a) notes that China’s regulatory landscape for digital governance can appear fragmented at times, with overlapping mandates among different governmental documents. Notably, there is a proliferation of “departmental rules” and “normative documents” issued by lower-level administrative bodies, which often carry significant *de facto* regulatory weight despite not being formal laws enacted by the legislature (Miao et al., 2021). Moreover, policies may be in conflict with formal law (Miao, 2024). For example, while rudimentary forms of real-name online registration emerged as early as 2012 (BBC News, 2012) and were later affirmed by a decision of China’s National People’s Congress (National People’s Congress Standing Committee, 2012), their explicit codification in formal law awaited the 2016 CSL. More recently, the mandate for displaying user IP addresses derives from a departmental rule issued by the CAC, situated two tiers below formal legislation. The nationwide health code system, implemented during the COVID-19 pandemic, while potentially drawing on high-level principles from laws concerning infectious disease prevention, was largely operationalized through policies developed by local governments.

Doubts also persist regarding whether the “collection and reporting of epidemic information” during emergencies strictly necessitates this specific format (Wang, 2020). Moreover, the CTID system itself, a representative case selected in this paper, is introduced via a normative document, underscoring the state’s capacity to effect significant privacy-related alterations through instruments of lesser formal legal standing.

The third is governance beyond law and policy. Under perceived conditions of political necessity, the party-state increasingly invokes a digitalized form of the “state of exception”, whereby legal norms are suspended or bypassed to advance social control and regime consolidation outside the formal boundaries of law and policy (Hu & Zhang, 2023). This manifests in two key dimensions. Substantially, state decisions that infringe upon individual privacy may lack any clear legal basis. For instance, in 2022, thousands of bank depositors had their health codes arbitrarily turned red to prevent them from traveling and protesting against frozen funds in rural banks (BBC News, 2022). Procedurally, there is a temporal disjunction between legal enactment and implementation. On the one hand, legal provisions may be applied retroactively to penalize behaviors that occurred before their official enactment—as in the 2022 penalty against Didi. On the other hand, legal drafts still at the consultation stage may already be implemented in practice, as illustrated by the early pilot rollout of the CTID Measures. To deflect mounting concerns about these systemic vulnerabilities, the Chinese party-state strategically cultivates an image as the primary defender of citizens’ privacy rights, particularly against abuses by private corporations, as identified by some scholars like Jia (2024) and Wang et al. (2024). This is achieved through a sophisticated interplay of policy pronouncements, legislative reforms, and highly publicized and selective enforcement actions (Jia, 2024). For instance, the 2022 penalty against Didi, though framed as a crackdown on unlawful data collection, served less to protect personal privacy—especially given that the company had previously been instructed to collect such data for public security purposes—and more to reassert state control over strategically sensitive data (Wang et al., 2024). This contrasts sharply with the state’s response to incidents like the reported Shanghai police database leak, which allegedly exposed records of over a billion citizens around the same period (Liu & Mozur, 2022). Unsurprisingly, however, information concerning the latter was heavily censored, and the public agency involved faced comparatively muted public accountability. Such selective enforcement and narrative management reinforce the state’s preferred image as a protector, rather than a potential violator, of privacy, thereby justifying its overarching control over data ecosystems (Wang et al., 2024).

In brief, the existing literature reveals a complex privacy landscape in China where *de jure* protections coexist with *de facto* vulnerabilities, particularly when individuals’ data interests intersect with state imperatives. The “authoritarian privacy” framework provides a crucial analytical lens for understanding how the lack of definitive constraints on state data practices, the instrumental use of diverse legal documents to enable privacy compromise, the existence of “state of exception” and the state’s narrative as a privacy guardian collectively contribute to the systemic vulnerability of individual online privacy. The following sections will demonstrate, through the Didi and CTID cases, how these dynamics operate in practice, legitimizing broader state objectives of data sovereignty and control.

3. Methodology

This study employs a case study approach integrated with qualitative content analysis to interrogate the structural vulnerabilities of privacy protection within China's evolving digital governance framework. The research focuses on two emblematic cases, *i.e.*, the Didi Global cybersecurity review case (2021-2022) and the implemented CTID system (2025), selected for their capacity to illuminate the legal and institutional mechanisms through which privacy vulnerabilities are institutionalized. These two cases are also emblematic in illustrating how, at critical junctures of rulemaking and enforcement, privacy language is invoked both to legitimize regulatory action and to expand state oversight. Moreover, the temporal scope of the cases (2021-2022) aligns with pivotal regulatory developments in China's digital society, including the enactment of PIPL and DSL, as well as the proliferation of subsidiary regulations governing cross-border data transfers. This period also marks the institutional consolidation of the CAC as the primary rule-maker and enforcer of digital governance, rendering the cases analytically salient for understanding the evolving legal landscape.

The Didi case, where Chinese cyberspace regulator issued by the time of writing the biggest fine according to several China's fundamental digital laws, represents a critical juncture in the enforcement of China's data privacy laws against private-sector actors. Initiated by a joint cybersecurity review investigation by multi regulators, this case ended with a huge emphasis on personal data privacy violations of the company while passing over the aspect of national data security in a few words. It can also be observed from this case how national security imperatives override corporate and individual privacy rights. Conversely, China's CTID system exemplifies state-driven initiatives that strategically deploy privacy rhetoric to legitimize expanded surveillance capabilities on the Internet. It also signals shift from segmental data controls toward a comprehensive identity architecture in China's digital society. Together, these cases reveal an important facet of China's digital governance that the normalization of state-centric governance objectives through ostensibly privacy-protective measures.

The data collection of this study specifically focuses on legal provisions and regulatory notices related to the two selected cases. These include the involved legal texts (*e.g.*, CSL, DSL, PIPL, the CTID Measures), case-specific government statements (*e.g.* government's announcement of a cybersecurity review on Didi, the official release of CTID system), and regulatory decisions (*e.g.*, CAC's Didi penalty notice). This method is designed to systematically explore large volumes of textual data, map underlying meanings, and generate critical understandings regarding how China posits privacy protection into its broader digital governance objectives. It primarily focuses on the interpretation and in-depth understanding of textual data, delving into contextual meanings and implicit messages. By limiting the analysis to legal texts, regulatory notifications, and government documents directly relevant to these cases, this study emphasizes the interplay between formal legal structures and their operationalization in China's digital governance practices.

The directed content analysis of this study is primarily structured around three dimensions. First, it examines how key statutory terms and provisions in China's digital laws are deliberately underspecified or overemphasized in each case. These terms, such as "privacy", "personal information", "national security", "data security", "public interest", and "cybersecurity", enable Chinese regulatory agencies to interpret and enforce in ways that balance or prioritize state governance objectives. Second, the analysis interrogates, in these

two representative cases, how the strategic application of privacy-related and security-related laws creates regulatory asymmetry between private and state actors in enforcement, such as the seemingly disproportionate penalty imposed on Didi under the PIPL's accountability framework. Third, it examines and critically analyses, under the authoritarian privacy theory proposed by Jia (2024), how broader governance narratives embedded in regulatory documents for the strategic invocation of legal terminology to legitimize individual privacy compromises.

It is, however, important to acknowledge certain limitations and delimitations of the methodology. The scope of this study is specifically delimited to a critical review of China's digital laws through the lens of two representative cases. While these cases are highly significant and illustrative, they do not encompass the entirety of China's vast and rapidly evolving digital governance landscape. The analysis focuses primarily on the legal and policy texts, official statements, and expert commentaries directly related to these cases, rather than extensive empirical data. This focused approach, however, aims to provide a critical review of legal frameworks and their practical implications within a specific analytical scope. Furthermore, the methodology acknowledges constraints in accessing non-public regulatory deliberations and opinions, particularly regarding the CAC's internal enforcement criteria. These data remain beyond the research's reach, potentially curtailing a fully comprehensive doctrinal mapping. Additionally, the CTID's recent implementation status necessitates caution in extrapolating its long-term governance implications.

4. The Didi Case: A Huge Penalty Overemphasizing Privacy Protection

The Didi case is a seminal event in China's rapidly evolving digital regulatory environment. The 8.026 billion Chinese yuan (approx. \$1.2 billion USD) fine levied against Didi Global Inc. by CAC on July 21, 2022, stands as the largest administrative penalty in China's data protection history. This decision followed a year-long probe initiated merely two days after Didi's June 2021 New York Stock Exchange (NYSE) initial public offering (IPO). The unprecedented fine signaled the Chinese government's resolve to vigorously enforce its new data security and personal information protection framework, particularly the PIPL and the DSL. Crucially, the investigation expanded beyond data privacy to encompass national security concerns and broader geopolitical aims, illustrating the intricate nexus of state control, economic strategy, and data governance in contemporary China.

4.1 Introduction: The Didi Case as a Nexus of China's Digital Governance

Didi was founded in 2012 and rapidly ascended to prominence as China's leading ride-hailing platform, as a disruptive force in the realm of urban mobility acclaimed for its innovative data-driven business model. However, Didi's extensive collection, storage, and utilization of vast quantities of personal and "important data"—ranging from sensitive individual consumer details like location and financial data to publicly critical information such as map data and traffic conditions—positioned it squarely within the crosshairs of China's burgeoning data protection framework. This operational paradigm, particularly its engagement with overseas financial markets, inherently intersected with China's escalating national data security and sovereignty concerns, resulting in a landmark regulatory intervention in China's digital governance.

The critical juncture that precipitates critical regulatory interventions in Didi's business was its IPO at the NYSE in June 2021 (Chen, 2021). Just two days later, the CAC, China's powerful

cyber regulator, launched an unprecedented cybersecurity investigation into Didi, citing potential national security risks linked to its large-scale data processing and cross-border data flows (CAC, 2021). This swift and severe response, which included on-site inspections in collaboration with other key ministries (public security, national security, transportation, *etc.*), indicated that the state's concerns transcended mere data privacy infractions but significant national security considerations and broader geopolitical objectives. The CAC also immediately issued a ban on Didi's mobile services, including the principal application "Didi Chuxing" and 25 associated applications, from Chinese app stores.

Table 1
A timeline of regulatory interventions in the Didi case

<i>Date</i>	<i>Didi Case Event</i>	<i>Concurrent Regulatory Development / Context</i>
30 June 2021	Didi Global Inc. listed on the New York Stock Exchange (NYSE)	
2 July 2021	CAC launched a cybersecurity review of Didi, citing national security risks	CAC announced the first public cybersecurity review under Cybersecurity Review Measures (CRM)
2 July 2021	CAC ordered Didi to suspend new user registration in China	
4 July 2021	CAC ordered removal of Didi apps from app stores due to illegal data collection	
10 July 2021	CAC proposed an updated version of CRM, requiring cybersecurity review for foreign listings by companies with >1M users	
1 September 2021		DSL took effect
1 November 2021		PIPL took effect
June 2022	Didi delisted from the NYSE	
21 July 2022	CAC announced ¥8.026 billion RMB (\$1.2 billion USD) fine on Didi Global Inc. and provided official explanation on the decision	Largest administrative fine in China's data protection sector
21 July 2022	Didi's CEO and President each fined RMB 1 million	

The year-long investigation concluded with the CAC citing a list of Didi's alleged contraventions of the PIPL's data protection requirements and briefly referring to Didi's violations under the DSL and the CSL, although the timing of investigation was even before the full effectiveness of the PIPL and the DSL (CAC, 2022b). The CAC's report characterized Didi's violations as "clear with solid evidence, serious circumstances and bad nature", meriting severe punishment (CAC, 2022a). As administrative sanctions of this scrutiny, the CAC resorted to a series of measures including mandates for remedial action and the imposition of substantial monetary penalties. Particularly, the monetary penalty against Didi, amounting to 8.026 billion Chinese yuan (around 1.2 billion USD), becomes the heaviest penalty that the CAC has issued in the name of data security so far (by the time of writing, *i.e.*, May 2025).

More or less, Didi's ambition for international capital was severely curtailed as its foreign IPO simultaneously raised alarm bells among Chinese regulators. The regulatory authorities perceived the listing process as a potential conduit for the outflow of sensitive personal and corporate data to foreign markets, causing broader uncertainties regarding and the firm's

commitment to conforming to domestic regulatory norms. Such a risk was also deemed inconsistent with the imperatives of national data security and sovereignty. Facing regulatory pressure and shareholder lawsuits (Bray, 2021), Didi announced its delisting from the NYSE in December 2021, completing it in June 2022, and signaled intentions for a Hong Kong listing. This outcome underscores the profound impact of China's politically infused regulatory landscape, where corporate actions are scrutinized through the lens of national strategy, and adherence to evolving, sometimes retroactively applied, data governance norms become paramount for survival and success.

4.2 Applicable Laws and Key Data Violations in Didi Case

The Didi case offers a significant lens through which to observe the real-world application and enforcement of China's pivotal digital laws. In July 2021, the CAC initiated a cybersecurity review of Didi's operations, invoking the National Security Law (NSL), the CSL and the CRM. Although the NSL does not expressly reference cybersecurity reviews, the mechanism is derived from its broader national security review framework under Article 59, which applies to activities—such as key technologies and network information services—that may impact national security. Under Article 35 of the CSL, operators of “critical information infrastructure”, a group of entities that are subject to heightened scrutiny due to their essential sectors, must undergo such reviews when procuring network products or services that could pose national security risks. Article 2 of the CRM further expands this obligation to include network platform operators' data processing activities that affect or may affect national security. Thus, the CAC's action against Didi can be viewed as a subset of the NSL's national security review regime, reflecting China's emphasis on safeguarding national security within cyberspace, particularly where data and digital platforms are concerned.

Despite “national security” being the initial and driving impetus for the cybersecurity review, the CAC's final decision on 21 July 2022 seemed to intentionally make it opaque regarding the specific nature of Didi's national security transgressions. The CAC vaguely concluded that Didi's data processing activities “seriously affected national security” and brought serious security risks to China's critical information infrastructure. Intriguingly, the announcement also highlighted Didi's broader illegal conduct, such as “refusing to comply with the clear requirements of regulatory authorities, paying lip service while secretly violating regulations, and maliciously evading supervision, which implies partially intrinsic philosophy of this biggest monetary fine in China's data governance. However, further details on both these critical findings were withheld, attributed to “national security considerations in accordance with law”.

The legal foundation for the substantial penalties against Didi rested primarily on three key legislations in China's data governance: CSL, DSL and PIPL. Although the PIPL, effective since 1 November 2021 as China's first comprehensive legislation for personal data protection, was not in effect when the investigation commenced, it became a central pillar of the CAC's final determination. While the CSL and DSL were cited, the CAC disproportionately emphasized PIPL violations, particularly focusing on breaches of consumers' personal data rights, to justify the severity of Didi's infractions. The unprecedented ¥8.026 billion RMB fine is widely believed by scholars to have been calculated based on the PIPL's liability clause (Shi et al., 2022), which permits penalties up to 5% of a company's preceding year's revenue under Article 66, though the CAC remained silent on the precise methodology of this calculation.

across the multiple laws.

The CAC detailed an extensive list of Didi's specific data violations, emphasizing severe infringements on user privacy and personal information rights under the PIPL. For example, Didi was accused of illegally and excessively collecting users' information, such as facial recognition data, precise location data and occupation information, and drivers' personal information, such as education information and ID numbers, going against the principles of lawfulness, necessity, legitimacy and minimization under Article 5 and 6 of the PIPL. Without fully performing its obligation of information notification under the PIPL (Article 17), Didi failed to expressly explain the purposes of collecting user information and illegally conducted data analytics on passengers' travel intentions, permanent cities and other related travel information. The CAC assessed these violations based on the degree of harm, the massive quantity of (sensitive) personal data involved (totaling 64.709 billion records), and the diversity of illegal activities, concluding that Didi's actions constituted a serious invasion of user privacy rights.

4.3 Invoking Privacy Language for Expanded State Oversight

The Didi case provides a striking example of the “authoritarian privacy” theory of Jia (2024), wherein autocratic states strategically invoke data-protection laws not primarily to safeguard individuals, but to bolster state authority and legitimacy. At the heart of the controversy lies the CAC's retroactive application of the PIPL: although the law only took effect on 1 November 2021, the CAC's final enforcement action reached back to June 2015, penalizing data-handling practices that predated the PIPL's existence. The PIPL was not referred in the beginning of CAC's cybersecurity review announcement but finally served as a strong legal basis of issuing severe penalties. Legally, this raises several thorny questions. Under both international norms and China's own Legislation Law, laws are ordinarily prohibited from applying to conduct that occurred before their effective dates—save in narrowly defined circumstances (e.g., for ongoing harmful activities). The CAC justified scrutinizing this entire seven-year period by arguing that Didi's violations persisted even after the PIPL's enactment. By doing so, the CAC effectively created an exception to the non-retroactivity principle, asserting that post-2021 continuations of pre-PIPL practices fell within the law's ambit.

This interpretive leap not only conflicts with the fundamental rule that new statutes should not criminalize or penalize past behavior, but also undermines administrative law safeguards against arbitrary governance. Critics argue contend that by sanctioning seven years of conduct—much of which was never subject to any binding data-protection regime—the CAC has sown legal uncertainty, leaving other technology companies unable to predict which legacy practices might suddenly be deemed unlawful (Zhang, 2022). In doing so, China risks establishing a “chilling precedent” for other technology firms (Zhang, 2022), whereby they must continually audit and police their historical archives against unforeseeable regulatory shifts, rather than relying on stable statutory guidance (Zhang 2022). More broadly, the Didi decision signals that China's data-privacy framework may be weaponized: instead of a predictable compliance landscape, businesses operate under the shadow of potential *ex post facto* enforcement, a dynamic that echoes the concerns central to the “authoritarian privacy” model.

Particularly, the CAC's official narrative surrounding the Didi case seems to deliberately overemphasize Didi's illegal activities under the PIPL, by means of highly critical wording,

overwhelming statistics and detailed violations. For example, the CAC described Didi's data practices as "extremely despicable" and underlined a seven-year history of non-compliance, including "feigning compliance" and "maliciously evading supervision". Its strong condemnation of Didi's "illegal collection of 64.709 billion personal data records" and the impact on "users' privacy rights" served to legitimize the severe administrative punishment, including the record fine under PIPL's liability provision and the unprecedented app removal and new user ban. This narrative also resonated with public concerns over data fraud and personal information misuse, allowing the regulator to harness public sentiment to support increased state intervention and oversight in digital governance. Although the government strategically presents itself as a responsive and responsible guardian of individual privacy (Wang et al., 2024), it is important to recognize the nature of this case as one of CAC's selective and symbolic enforcement efforts, where personal privacy infringement might not have been the solo motivation behind its swift actions.

Indeed, the Didi case crystallized China's intensified focus on data as a national security asset beyond Chinese citizens' privacy, revealing the growing tensions between important data, data localization, and national security. CAC's stringent scrutiny this case further confirms that Didi is classified as a "critical information infrastructure operator" (CIIO) under the CSL and the DSL, a group of entities subject to stringent data localization requirements and rigorous security assessments for cross-border transfers, due to its vast data holdings (*e.g.*, geographic, economic, personal data) (Dorwart & Zafir-Fortuna, 2021). Didi's overseas IPO further magnify such concerns due to the potential exposure of company data to US regulators, while specific national security breaches remained undisclosed in the decision. It is thus unsurprising that the CAC swiftly revised the CRM, mandating cybersecurity reviews for platform operators with over one million users before overseas listings, which further fosters data localization mandate for CIIOs. In this sense, it would also be arguably that this regulatory crackdown also advanced broader geopolitical and economic policy goals, such as reinforcing China's "digital sovereignty" by controlling data and capital outflows and incentivizing domestic IPOs.

As Zhang (2022) argues, the Didi enforcement action is also an example proving the significant expansion of CAC's institutional power and its dominion over the digital economy. Its swift and decisive actions, including the app removals and new user registration bans imposed even when the investigation was launched, demonstrated its enforcement capabilities in China's digital society and further solidified its status as a "super regulator". Even though before the Didi case, the CAC had been already seen as China's super cyber regulator, being a party-state entity characterized by a lack of institutional transparency and accountability typical of other administrative agencies (Horsley, 2022; Ma & Liu, 2024). Such opaque and politically connected nature also allows the CAC operational flexibility and discretion in deciding its regulatory narrative in Didi case. Although there is an unproportionate mentioning on personal data infringements in the CAC's investigation decision, it far transcends traditional notions of privacy enforcement because of broader governance, geopolitical and economic objectives, being a highly visible and punitive demonstration designed to deter other companies from approaching the intangible highwire of national data security.

5. The CTID System: A Tool That Could Enhance State-Centric Surveillance

As China deepens its digital governance ambitions, the CTID system has emerged as a critical infrastructure for centralizing online identity verification. While officially introduced as a tool

to enhance online privacy and strengthen cybersecurity, the CTID system in practice will reinforce centralized state control over digital identities. This section critically examines its legal foundations, operational logic, and privacy implications of the CTID system. It demonstrates that the system is likely to be a toolkit for reinforcing digital surveillance.

5.1 Introduction: The Emergence of a Unified Digital Identity and Its Legal Foundations

The CTID system represents a pivotal development in China's strategy for online authentication, cybersecurity, and data governance. Derived from the existing national ID card infrastructure (ANICERT, n.d.), it transcends the concept of a mere electronic counterpart to physical identification. Instead, it advances what official promulgations term "the trusted online identity strategy" (Article 1 of the CTID Measures), positioning it as an autonomous mechanism for verifying individual identities in the digital realm. While officially framed as a measure to bolster cybersecurity and enforce data protection, the CTID system engenders substantial concerns regarding the safeguarding of personal data and individual privacy within China's extant legal framework (BBC, 2024).

The CTID Measures purport to draw their legal basis from several superordinate statutes. Article 24 of the CSL is invoked to support a network identity credibility strategy and the development of secure electronic identity authentication technologies. Similarly, Article 33 of the DSL and Article 62 of the PIPL mandate that data transaction intermediary institutions verify data origins and transaction parties' identities, maintaining records of these processes, coupled with obligations to maintain records of these verification processes. Further support is cited from Article 33 of the Law on Countering Telecommunications Network Fraud and Article 75 of the Law on the Protection of Minors, which authorities online intermediaries to verify the identity of accounts to verify the identities of users suspected of fraud or minors engaging with electronic games.

However, the objective of online identity verification does not intrinsically necessitate the creation and application for a distinct Network Certificate or Network Number as envisaged by the CTID Measures. At its core, the CTID system, much like pre-existing real-name registration mandates, is centered on authentication—the process of ensuring an individual's online persona corresponds to their verifiable real-world identity. Objectives such as preventing online platforms from excessive collection or leakage of personal information can be, and are, addressed through existing regulatory frameworks. For instance, the prevailing national ID system, in conjunction with real-name registration requirements for telephone and network access services, already furnishes a robust infrastructure sufficient to meet most online identity verification needs. Consequently, the specific provisions within the CTID Measures compelling "applications for Network Numbers or Network Certificate" appear to lack a firm grounding in higher-level legislation. This requirement, central to the operationalization of the CTID system, seems to be a new mandate originating from a set of administrative measures rather than an initiative clearly rooted in, or necessitated by, established statutory authority. This raises questions about the proportionality and necessity of imposing a new, overarching digital identity layer.

5.2 Comparative Analysis: Toward a More Centralized and Expansive System

Compared with previous systems like real-name registration and health codes deployed by

Chinese government, the CTID system reflects a distinct trend toward greater centralization of data control and a broader scope of application. Real-name registration, while universal across various facets of internet access and usage in China, operates on a distributed model where the burden of identity verification and data retention falls upon individual service providers (Lee & Liu, 2016, Hu, 2013). Conversely, the health code system, though largely confined to the public health domain during the COVID-19 pandemic, involved data collection and control highly centralized under state authorities. The CTID system combines both characteristics: it is both comprehensive in its intended applicability across digital services and centrally managed, effectively positioning the state as a “super” data controller and processor.

Specifically, this structural shift represents a significant departure from the previous real-name registration regime, while they share one feature of eliminating cyberspace anonymity (Lee & Liu, 2016). Under the latter, platforms are individually mandated to collect and store user identity information, with the state exercising oversight “at a distance” by disciplining users through the platforms themselves (Jiang, 2016). This resulted in a fragmented data landscape where government access is typically indirect, mediated through requests to these disparate platform holders. In stark contrast, the CTID system is designed to centralize identity verification under the unified management of the MPS and the CAC. Users would be assigned a unique, cross-platform digital identifier, with their identity data either centrally stored or directly managed by a public service platform under state control. This shift does not inherently reduce identity verification or data collection; rather, it reallocates primary responsibility for these functions from private platforms to state entities. The real-name registration system has already evidenced how the integration of administrative and technological architectures can intensify Internet control (Tu, 2024), a trend the CTID system appears ready to accelerate.

The operational logic of the CTID system also shares strong parallels with the health code system deployed during the COVID-19 pandemic. As noted by commentators such as Lao (2024), both systems share a common rationale and implementation model, while the CTID system carries these control mechanisms appeared in the pandemic into the realm of normalized, long-term governance. . The health code system integrated personal identity data (national ID numbers, phone numbers), health information (body temperature, test results), and geolocation data to generate dynamic risk assessments, enabling local governments—under national coordination—to conduct wide-scale surveillance of individuals’ movements and health status (Huang et al., 2022) for purposes of extensive social control (Cong, 2021). The CTID system, while focused on identity verification, shares key characteristics: it enables comprehensive monitoring of digital activities by linking them to a verified identity, and it is state driven, relying on centralized national databases for verification. This reflects a broader strategic ambition to implement digital governance through integrated data infrastructures. The health code, originally framed as a temporary emergency measure, has shown tendencies towards integration into long-term governance strategies (Cong, 2021). The CTID system represents a more expansive and potentially permanent endeavor to embed state-mediated identity verification deep within the digital society.

5.3 Privacy Vulnerabilities under State-Centered Governance

According to CAC’s explanations on the CTID Measures, the centralized digital identity management by the state can “minimize the excessive collection and retention of personal data by online platforms under the pretext of complying with the real-name registration

requirement”. This narrative, consistent with other internet-related legislative efforts in China, positions the party-state as a privacy guardian against potential overreach by the private sector (Wang et al., 2024).

However, this assumption warrants critical examination. The assertion that government-led data collection is inherently more secure or trustworthy than platform-based collection is contestable, as both private and state actors in practice inherently carry risks of excessive collection, misuse, and even data manipulation and surveillance (Richards, 2013, Altman et al. 2018). While the CTID system purports to limit data aggregation by corporations, it simultaneously consolidates a vast repository of sensitive identity information under direct state control. This centralization raises concerns about potential government overreach in the absence of transparency and meaningful consent the intensification of state surveillance capabilities, ultimately rendering individual privacy more vulnerable..

5.4 Challenges in Transparency and Consent

Transparency and informed consent are essential aspects of privacy protection regimes (Taylor, 2017), yet both remain fragile in CTID’s practice. Transparency requires clear, accessible communication about who processes personal data, for what purpose, and ensures individuals can understand and access their data (Article 29 Data Protection Working Party, 2018). The CTID system, however, may not meet these standards, as it offers only limited and ambiguous explanations about the scope of data collection, the modalities of processing, and the entities with access, with exception (Article 11 of the CTID Measures). It also fails to clarify whether and how such data may be shared or further utilized by authorities within China’s centralized political framework, as the CAC and other relevant authorities serve as the competent supervisory bodies (Article 3 of the CTID Measures).

Closely tied to the issue of transparency is the question of whether individuals can genuinely exercise informed consent in the data collection process. Digital identity systems like CTID involve large-scale processing of personal data, including sensitive biometric information such as facial recognition data. For instance, applying for a Network Number or Network Certificate requires ID documents, facial scans, and phone numbers. While Article 13 of the PIPL mandates consent for processing personal data, it also contains broad exceptions for processing activities related to governmental duties, potentially allowing state authorities to bypass explicit consent requirements. The CTID Measures themselves, while stipulating that public service platforms must fulfill notification obligations when processing personal and sensitive personal information, incorporate sweeping exceptions in Articles 11: “except as otherwise stipulated by laws” and “situations that laws or administrative regulations provide shall be kept confidential”. This provision creates a significant lacuna potentially enabling state misuse of personal information collected through the CTID system under vaguely defined legal or regulatory justifications.

Furthermore, the absence of robust, independent data protection oversight in China critically exacerbates these privacy risks. Within the CTID architecture, government bodies such as the MPS function concurrently as data controllers, determining the purposes (e.g., cybersecurity enhancement) and means (e.g., biometric authentication) of data processing, and directly undertaking the collection and management of this data. Crucially, China’s current legal and institutional landscape lacks a genuinely independent body capable of effectively supervising the state’s own data collection and processing activities. This institutional feature has a

significant impact on effective redress mechanisms: individuals encountering privacy violations or seeking to exercise their data rights—such as access, correction, or deletion—within the CTID system may face huge challenges in obtaining effective remedies.

5.5 Intensified Digital Surveillance

Another privacy-related issue is that the data collected and processed by the CTID system might unlikely be confined strictly to the stated purpose of identity authentication, such as extending to encompass users' online behaviors and digital footprints, thereby enabling pervasive digital surveillance (Tobin et al., 2024). This apprehension is rooted in the system's centralized architecture, combined with the extensive legal authority granted to government agencies to access data for broadly defined purposes such as national security or public interest, as granted by laws like the PIPL and the CSL. This raises critical questions regarding the ultimate scope of monitoring: could the system be leveraged to track online activities, associate them with real identities, map physical movements, or even integrate with other state-led initiatives like the social credit system?

Such concerns are not unfounded. Evidence suggests state's capabilities will go beyond simple identity verification as the public service platform is reported to be providing "online behavior tracing" services, beyond online identity management and real identity verification (ANICERT, n.d.). This expansion significantly broadens the scope of surveillance. It remains unclear under what conditions such tracing is triggered, for what purposes it is carried out, and what consequences may arise for individuals whose online behavior is deemed sensitive or problematic. The introduction of "online behavior tracing" could plausibly indicate that information beyond identity—such as online speech and activity—might be subject to increased monitoring. What's more, digital surveillance enables authorities to track political dissidents through technology-driven mechanisms (Hu & Zhang, 2023). Such data monitoring significantly undermines user anonymity online (Lee & Liu, 2016) and exert a chilling effect on individuals' freedom of expression, especially when engaging in discussions on political or socially sensitive topics. Fearing identification and state scrutiny, users may increasingly resort to self-censorship (Penny, 2017, Büchi et al., 2022). These behavioral adaptations are highly likely to intensify under the CTID regime. Ultimately, the system may not merely inhibit open discourse but also contributes to a broader contraction of civic space, where dissenting opinions are marginalized and public expression becomes more restrained—thereby reinforcing a system of pervasive digital surveillance.

6. Conclusion

This study has shown how China's invocation of personal privacy protection serves less as a safeguard for individuals and more as a strategic tool for expanding state oversight. The Didi case exemplifies this "authoritarian privacy" strategy: by retroactively applying the privacy-related legal provisions and deploying strong rhetoric around personal data abuses, the CAC strengthened its regulatory remit, heightened enforcement uncertainty, and reinforced data-sovereignty imperatives under the guise of protecting users. Likewise, the CTID system crystallizes a broader shift toward state-centralized digital identity management, merging features of real-name registration and pandemic health codes, to recast the state as the ultimate data controller. Although presented as a means to curb private-sector data overcollection, CTID concentrates sensitive biometric and behavioral data within public authorities, potentially undermining transparency, consent, and independent oversight and amplifying

pervasive digital surveillance. Together, these cases reveal a consistent pattern: the rhetoric of privacy protection is instrumentalized to justify deeper governance, security, and geopolitical objectives, even as individual autonomy and legal safeguards recede.

The Chinese privacy regime sheds critical light on broader debates in global privacy governance by revealing how privacy can be mobilized not to limit state power, as in democracies, but to reinforce it. In contrast to Western models—where privacy protection often functions as a constraint on both state and corporate actors—China’s approach illustrates how privacy discourse can be appropriated to legitimize extensive data collection, centralized identity control, and pervasive digital surveillance. This inversion of privacy’s normative role underscores the need for more context-sensitive analyses of data governance regimes, particularly in authoritarian contexts where superficial privacy protections may coexist with expansive state prerogatives.

Future research may extend beyond privacy to examine how digital authoritarianism manifests across broader governance domains, such as content regulation and algorithmic control. As this study suggests, privacy protections in authoritarian regimes often serve governance imperatives rather than individual rights, highlighting the need for conceptually sensitive comparative approaches. At the same time, democracies must remain vigilant against the instrumentalization of privacy laws—ensuring they serve as meaningful safeguards rather than symbolic gestures. These questions call for deeper inquiry into how legal frameworks, across regimes, can uphold substantive privacy protections in the digital era.

References

- Altman, M., Wood, A., O'Brien, D. R., & Gasser, U. (2018). Practical approaches to big data privacy over time. *International Data Privacy Law*, 8(1), 29–51.
<https://doi.org/10.1093/idpl/ix027>
- ANICERT. (n.d.). *CTID platform introduction*. Retrieved May 28, 2025, from <https://www.anicert.cn/platform.html?md=2>
- Article 29 Data Protection Working Party. (2018, August 22). *Guidelines on transparency under Regulation 2016/679* (wp260rev.01). European Commission.
<https://ec.europa.eu/newsroom/article29/items/622227>
- BBC. (2024, August 6). *China begins trial operation of “online identity” and “online number” systems; Netizens worry online rights “will be stripped”*.
<https://www.bbc.com/zhongwen/simp/chinese-news-69244432>
- BBC News. (2012, March 15). *China’s Weibo begins blocking posts from unverified users*.
https://www.bbc.com/zhongwen/simp/chinese_news/2012/03/120315_weibo_name
- BBC News. (2022, June 14). *Henan: China Covid app restricts residents after banking protests*. <https://www.bbc.com/news/world-asia-china-61793149>
- Bray, C. (2021, July 8). Didi Chuxing sued by American shareholders over stock plunge caused by Chinese regulatory changes, data-collection inquiry. *South China Morning Post*. <https://www.scmp.com/business/banking-finance/article/3140272/didi-chuxing-sued-american-shareholders-over-stock-plunge>
- Büchi, M., Festic, N., & Latzer, M. (2022). The chilling effects of digital dataveillance: A theoretical model and an empirical research agenda. *Big Data & Society*, 9(1), 1–14.
<https://doi.org/10.1177/20539517211065368>
- Chen, Q. (2021, July 1). Didi raises \$4.4 billion in largest China-US IPO since Alibaba. *TechNode*. <https://technode.com/2021/07/01/didi-raises-4-4-billion-in-largest-china-us-ipo-since-alibaba/>
- Chengdu Commercial Daily. (2024, August 3). Curbing platforms from excessively collecting personal information, "Online ID card" is coming. CCTV.com.
<https://news.cctv.com/2024/08/03/ARTI1qZmVcMexWvMfBC1P6IN240803.shtml>
- Cong, W. (2021). From pandemic control to data-driven governance: The case of China’s health code. *Frontiers in Political Science*, 3, 627959.
<https://doi.org/10.3389/fpos.2021.627959>
- Creemers, R. (2022). China’s emerging data protection framework. *Journal of Cybersecurity*, 8(1), tyac011. <https://doi.org/10.1093/cybsec/tyac011>
- CAC. (2020, February 16). *National version of health code to go online next week*.
https://www.cac.gov.cn/2020-02/16/c_1583402314064163.htm
- CAC. (2021, July 2). *Notice by the Cybersecurity Review Office on initiating a cybersecurity review of “Didi Chuxing”*. https://www.cac.gov.cn/2021-07/02/c_1626811521011934.htm
- CAC. (2022a, July 21). *CAC official answers questions on administrative penalty decision related to cybersecurity review of Didi Global Inc*. https://www.cac.gov.cn/2022-07/21/c_1660021534364976.htm
- CAC. (2022b, July 21). *Decision on administrative penalty related to cybersecurity review made in accordance with law against Didi Global Inc*. https://www.cac.gov.cn/2022-07/21/c_1660021534306352.htm
- CAC. (2022c, August 1). *Management of Internet user account information*.
https://www.cac.gov.cn/2022-08/01/c_1660644109273814.htm

- CAC. (2022d). *Cybersecurity review measures* [Effective February 15, 2022]. Retrieved from http://www.cac.gov.cn/2022-01/04/c_1642894601639024.htm
- CAC. (2023, July 10). *Notice on strengthening the management of “personal media”*. http://www.cac.gov.cn/2023-07/10/c_1690638496047430.htm
- De Hert, P., & Papakonstantinou, V. (2015). The data protection regime in China. *Depth Analysis//Brussels Privacy Hub Working Paper*, 1(4).
- Dorwart, H., & Zafir-Fortuna, G. (2021, July 23). *Spotlight on the emerging Chinese data protection framework: Lessons learned from the unprecedented investigation of Didi Chuxing*. Future of Privacy Forum. <https://fpf.org/blog/spotlight-on-the-emerging-chinese-data-protection-framework-lessons-learned-from-the-unprecedented-investigation-of-didi-chuxing/>
- Gorman, L. (2021). China’s data ambitions: Strategy, emerging technologies, and implications for democracies. *National Bureau of Asian Research*.
- He, A. (2023). *State-centric data governance in China* (CIGI Papers No. 282). Centre for International Governance Innovation. <https://hdl.handle.net/10419/299979>
- Horsley, J. P. (2022, August 8). *Behind the facade of China’s cyber super-regulator*. DigiChina. <https://digichina.stanford.edu/work/behind-the-facade-of-chinas-cyber-super-regulator/>
- Hu, H. L. (2013). Real name systems in Chinese cyberspace: Authentication, privacy, and state capacity. *Peking University Journal of Legal Studies*, 4(1), 207–232.
- Hu, J., & Zhang, X. (2023). Digital Governance in China: Dispute Settlement and Stability Maintenance in the Digital Age. *Journal of Contemporary China*, 33(148), 561–577. <https://doi.org/10.1080/10670564.2023.2261877>
- Huang, G., Hu, A., & Chen, W. (2022). Privacy at risk? Understanding the perceived privacy protection of health code apps in China. *Big Data & Society*, 9(2), 20539517221135132.
- Jaguścik, J., & Imbach, J. (2022). Digital society in China. In *Asiatische Studien-Études Asiatiques* (Vol. 76, Issue 1, pp. 1–9). De Gruyter.
- Jia, M. (2024). Authoritarian Privacy. *University of Chicago Law Review*, 91, 733–809. <https://doi.org/10.2139/ssrn.4362527>
- Jiang, M. (2016). Managing the micro-self: The governmentality of real name registration policy in Chinese microblogosphere. *Information, Communication & Society*, 19(2), 203–220. <http://dx.doi.org/10.2139/ssrn.2436922>
- Lao, D. (2024, July 30). “Online number and online certificate” is a monitor installed on every netizen [Archived version]. *China Digital Times*. <https://chinadigitaltimes.net/chinese/710151.html>
- Lee, C. S. (2019). *Datafication, dataveillance, and the social credit system as China’s new normal*. *Online Information Review*, 43(6), 952–970. <https://doi.org/10.1108/OIR-08-2018-0231>
- Lee, J.-A., & Liu, C.-Y. (2016). Real-name registration rules and the fading digital anonymity in China. *Wash. Int’l LJ*, 25, 1. Available at: <https://digitalcommons.law.uw.edu/wilj/vol25/iss1/3>
- Li, T. C., Bronfman, J., & Zhou, Z. (2017). Saving face: Unfolding the screen of Chinese privacy law. *Journal of Law, Information, and Science (Forthcoming)*. Available at: https://scholarship.law.bu.edu/faculty_scholarship/816
- Liu, J., & Mozur, P. (2022, July 5). Hackers claim theft of police info in China’s largest data leak. *The New York Times*. <https://www.nytimes.com/2022/07/05/business/china-police-data-breach.html>

- Ma, Y., & Liu, C. (2024). The developmental party and the regulatory state in China's Internet governance. *Policy & Internet*, 16(4), 764–782.
<https://doi.org/10.1002/poi3.410>
- Miao, W., Jiang, M., & Pang, Y. (2021). Historicizing internet regulation in China: A meta-analysis of Chinese internet policies (1994–2017). *International Journal of Communication*, 15, 24.
- Miao, M. (2024). *Coded social control: China's normalization of biometric surveillance in the post COVID-19 era*. *Washington Journal of Law, Technology & Arts*, 19(1).
<https://digitalcommons.law.uw.edu/wjlta/vol19/iss1/2>
- Ministry of Public Security, Cyberspace Administration of China, Ministry of Civil Affairs, Ministry of Culture and Tourism, National Health Commission, & National Radio and Television Administration. (2025, May 23). *Measures on the Management of the National Online Identity Authentication Public Service*.
https://www.cac.gov.cn/2025-05/23/c_1749711107835487.htm
- National People's Congress. (2016). *Cybersecurity Law of the People's Republic of China*.
http://www.npc.gov.cn/zgrdw/npc/xinwen/2016-11/07/content_2001605.htm
- National People's Congress. (2021, June 11). *Data Security Law of the People's Republic of China*. https://www.gov.cn/xinwen/2021-06/11/content_5616919.htm
- National People's Congress. (2021, August 20). *Personal Information Protection Law of the People's Republic of China*. http://www.gov.cn/xinwen/2021-08/20/content_5632486.htm
- National People's Congress Standing Committee. (2012, December 28). *Decision of the Standing Committee of the National People's Congress on strengthening the protection of online information*. https://www.gov.cn/jrzg/2012-12/28/content_2301231.htm
- Penney, J. W. (2017). Internet surveillance, regulation, and chilling effects online: A comparative case study. *Internet Policy Review*, 6(2).
<https://doi.org/10.14763/2017.2.692>
- Pernot-Leplay, E. (2020). China's Approach on Data Privacy Law: A Third Way Between the U.S. and the E.U.? *International Affairs*. Available at SSRN: <https://ssrn.com/abstract=3542820>
- Richards, N. M. (2013). The dangers of surveillance. *Harvard Law Review*, 126(7), 1934–1965.
- Roberts, H. (2022). Informational privacy with Chinese characteristics. In *The 2021 Yearbook of the Digital Ethics Lab* (pp. 9–23). Springer.
- Shi, M., Horsley, J. P., & Lu, X. (2022, July 22). *Unpacking the DiDi decision*. DigiChina, Stanford University. <https://digichina.stanford.edu/work/forum-unpacking-the-didi-decision/>
- Taylor, R. (2017). No privacy without transparency. *Data Protection and Privacy: The Age of Intelligent Machines, Computers, Privacy and Data Protection*, 63–88.
- Tobin, M., et al. (2024, July 31). China wants to start a national internet ID system. *The New York Times*. <https://www.nytimes.com/2024/07/31/business/china-national-internet-id.html>
- Tu, M. (2024, April 17). *Weibo frontend real-name system: Further tightening of speech space and Up hosts' fear of being "doxed"*. WAINAO.

- <https://www.wainao.me/wainao-reads/weibo-real-name-policy-speech-user-privacy-04172024/>
- Wang, M. (2020, April 1). China: Fighting COVID-19 with automated tyranny. *The Diplomat*. <https://thediplomat.com/2020/03/china-fighting-covid-19-with-automated-tyranny/>
- Wang, R., Zhang, C., & Lei, Y. (2024). Justifying a Privacy Guardian in Discourse and Behaviour: The People's Republic of China's Strategic Framing in Data Governance. *The International Spectator*, 59(2), 58–76. <https://doi.org/10.1080/03932729.2024.2315064>
- Yang, Z. (2023, December 22). *The death of anonymity online in China*. MIT Technology Review. <https://www.technologyreview.com/2023/12/22/1085820/death-of-anonymity-online-china/>
- Zhang, A. H. (2022, September 20). *The Didi case and the Party's influence over data enforcement* (Vol. 3, No. 2). USALI Perspectives. <https://usali.org/usali-perspectives-blog/the-didi-case-and-the-partys-influence-over-data-enforcement>
- Zhang, C. (2024a). China's privacy protection strategy and its geopolitical implications. *Asian Review of Political Economy*, 3(1), 6. <https://doi.org/10.1007/s44216-024-00028-2>



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